

Teamwork Agreement Template

For the MathWorks Classroom Challenge Projects

Project: Drone Payload Capacity and Structural Design Analysis

Team 5

Members and Roles

1. Chris Penick: Design + CAD Lead
2. Edwin Chen: FEA + Structural Safety Lead
3. Juan Gonzales: Project Manager + Report/QA Lead
4. William McMurray: Thrust-to-Weight + Payload Analysis Lead

1) Purpose & Scope

We agree to collaborate to deliver a high-quality, ethical, and reproducible project. Our work will follow the project guidance and produce the specified artifacts (e.g. plots, code, and visualizations).

2) Roles & Responsibilities

Person 1 — Project Manager + Report/QA Lead

Responsible for:

- Scheduling team meetings
- Maintaining the timeline and task board
- Organizing the shared folder or GitHub/Google Drive
- Creating the report template
- Writing the problem statement, assumptions, limitations, and next steps
- Checking units, formatting, figure labels, table labels, and file organization
- Re-running final MATLAB files to check reproducibility
- Combining everyone's sections into the final report

Person 2 — Design + CAD Lead

Responsible for:

- Creating two meaningfully different drone arm designs
- Producing clear labeled sketches
- Defining dimensions, motor mount location, and body connection
- Creating CAD models for both designs
- Exporting STEP/STL files for FEA
- Providing design screenshots and descriptions for the report
- Helping the FEA lead identify correct faces for supports and loads

The project specifically requires sketches showing geometry, dimensions, motor mount location, and where the arm connects to the drone body.

Person 3 — Thrust-to-Weight + Payload Analysis Lead

Responsible for:

- Loading material properties into MATLAB
- Organizing material data into a table
- Calculating arm volume and mass for each design/material option
- Calculating total drone mass
- Applying the 2:1 thrust-to-weight requirement
- Calculating maximum payload capacity
- Creating the 12-case results table: 2 designs × 6 materials
- Creating payload comparison plots
- Writing the thrust-to-weight analysis section

Person 4 — FEA + Structural Safety Lead

Responsible for:

- Importing CAD models into MATLAB
- Applying boundary conditions
- Applying upward motor thrust and downward motor weight
- Running FEA for each design/material combination
- Reporting maximum displacement, maximum von Mises stress, and factor of safety
- Creating x-, y-, z-displacement and von Mises stress plots
- Writing the structural safety/FEA section

3) Communication Norms

- Primary channel: Discord
- Secondary channel:
 - Will - (510) 605 8935
 - Chris - (408) 888 0683
 - Juan- 909 212 9837
 - Edwin - (626) 235-2784
- Response time: 24 hrs on weekdays and 48 hrs on weekends
- Meeting cadence: __weekly working meeting (60–90 minutes on Wednesdays via Zoom at 5:15-5:30)

- Decision records:
 - Person 1 will keep a running decision log in the shared folder.
 - Major decisions must include the date, decision made, reason, and who agreed.

4) Collaboration & Tools

- Version control: GitHub folder
- Data management: All datasets, parameters, and assumptions documented in

README.md

File naming & style: Use clear descriptive names with version numbers.

Example:

- Design_A_Rectangular_Beam_v1.step
 - Design_B_Truss_Arm_v1.step
 - Thrust_to_Weight_Analysis_v1.mlx
 - FEA_Results_Design_A_v1.mlx
 - Final_Report_v1.docx
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- Backups: Person 1 will keep old versions in an “Archive” folder. No one deletes another teammate’s work without asking.

5) Quality Standards (Definition of 'Done')

- ☐ It meets the project requirements.
- ☐ Units are included and consistent.
- ☐ Assumptions are documented.
- ☐ Figures and tables have titles, labels, and units.
- ☐ MATLAB code runs without missing files.
- ☐ CAD files open correctly and can be imported/exported.
- ☐ Results are checked by at least one teammate.
- ☐ The final recommendation is supported by quantitative results.

- ☐ All files are saved in the shared folder.

6) Timeline & Milestones

Week 1: Setup, assumptions, and design direction

- Person 1 creates a shared folder, task board, report template, and assumptions table.
- Person 2 sketches two arm designs and labels dimensions.
- Person 3 loads material properties and starts MATLAB setup.
- Person 4 reviews the FEA workflow and tests geometry import.

Week 1 goal:

Two designs selected, assumptions documented, shared folder organized, and MATLAB/FEA workflow started.

Week 2: CAD models and first calculations

- Person 1 reviews formatting, file names, and assumptions.
- Person 2 finishes CAD models and exports STEP/STL files.
- Person 3 completes first thrust-to-weight table.
- Person 4 runs one test FEA case.

Week 2 goal:

CAD files completed, first payload table completed, and one FEA test case working.

Week 3: Complete analysis

- Person 1 checks tables, figures, and units.
- Person 2 fixes CAD issues and prepares final design visuals.
- Person 3 finalizes all 12 payload capacity results.
- Person 4 completes FEA results for all design/material combinations.

Week 3 goal:

Final payload analysis and FEA results completed.

Week 4: Final report and recommendation

- Person 1 assembles and formats the final report.
- Person 2 finalizes design description and visuals.
- Person 3 finalizes payload analysis discussion.
- Person 4 finalizes FEA discussion.
- Everyone reviews the final report and prepares for presentation if required.

7) Decision-Making & Conflict Resolution

- Default: _We will aim for consensus after discussion
- If blocked >48 hrs: _The team will hold a short discussion, compare evidence, and take a majority vote.
- Technical disagreements: _We will compare options using project requirements, calculations, FEA results, payload capacity, factor of safety, cost, and practicality.
- Escalation: If unresolved, consult instructor/mentor with a concise memo.

8) Workload, Accountability & Attendance

Task board: Trello

We will use a shared task board with four columns:

- To Do
- Doing
- Review
- Done

Each task must have:

- Owner
- Due date
- Status
- Reviewer

Accountability:

If someone misses a deadline, they must provide a recovery plan within 24 hours. If deadlines are repeatedly missed, the team may redistribute tasks.

Attendance:

If someone will miss a meeting, they must give notice and post their progress update before the meeting.

10) Accessibility & Inclusion

We commit to respectful collaboration, equitable turn-taking, and flexibility for documented accessibility needs.

11) Deliverables Checklist

- ☐ Two labeled drone arm sketches
- ☐ Two CAD arm models
- ☐ STEP/STL files for FEA
- ☐ Material properties table
- ☐ Assumptions table
- ☐ MATLAB thrust-to-weight analysis
- ☐ 12-case payload capacity results table
- ☐ Payload comparison plot
- ☐ MATLAB FEA analysis
- ☐ Stress, displacement, and factor of safety table
- ☐ x-, y-, z-displacement plots
- ☐ Von Mises stress plots
- ☐ Final recommended design solution

☐ Final report

12) Agreement & Signatures

Name & Signature: _____Edwin Chen_____ Date:

__07/06/2026__

Name & Signature: _____William McMurray_____

Date: __07/06/2026__

Name & Signature: _____Chris Penick_____ Date:

__07/06/2026__

Name & Signature: _____Juan Gonzales_____ Date:

__07/06/2026__